



UNIVERSITY CATEGORY

REGISTRATION FORM ROUND 1

BACH KHOA INNOVATION 2026

1. **TEAM'S NAME:** Sankit - Next-Gen Farming

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PROJECT'S INFORMATION

1. PROJECT'S NAME: Sankit - Next-Gen Farming: Quality Evidence Platform for Medicinal Herb Supply Chains
• <i>Computer Science and Engineering;</i> • <i>Environment and Resources;</i> • <i>Industrial Management, Business Administration;</i>
2. SUMMARIZING YOUR IDEA/PROJECT (Maximum of 500 words) Sankit is a digital platform that helps medicinal herb farms, cooperatives, and enterprises build reliable proof of quality throughout the cultivation process. The core problem in the medicinal herb market is not simply the lack of farming apps, QR codes, sensors, or traceability tools. These technologies already exist. The deeper problem is that many farms still cannot prove, in a continuous and verifiable way, how a medicinal herb batch was grown, monitored, harvested, and prepared. This matters because medicinal herbs are not valued only by yield or appearance. Their quality depends on plant identity, growing area, soil and water conditions, microclimate, cultivation practices, harvest timing, primary processing, storage, and consistency across seasons. WHO's GACP guidance also emphasizes that quality assurance for medicinal plant materials starts from cultivation and collection, not only from later manufacturing. (WHO, 2003) In practice, farm records are often scattered across paper notebooks, Excel files, messaging apps, separate photo folders, and documents prepared only before inspections. Generic digital agriculture tools often fail to scale because they create extra data-entry work for field users, do not clearly translate into business value, and are usually sold to the wrong payer. Sankit is designed around this adoption reality: the field worker may be the daily user, but the real economic buyer is often the farm owner, cooperative, processor, medicinal herb enterprise, or buyer that needs reliable quality records. Sankit starts with a focused product package: the GACP Readiness and Farm Trust Pack. It combines an offline-first mobile field app, QR/GPS plot identification, timestamped photo and video evidence, task workflows, plan-versus-actual monitoring, basic sensor integration, risk alerts, and audit-ready reporting. Over time, Sankit develops three higher-value layers: a Quality Formation Passport for each batch, a Quality Trust Score for farms and suppliers, and an AI Quality Risk Engine to detect missing evidence, abnormal field conditions, and process risks under expert review. Blockchain is used only to protect verified milestones after approval, not to claim that raw field data is automatically true. Sankit is not another farm diary app. It is a practical system for turning cultivation activities into trusted quality proof

3. DETAILS OF YOUR IDEA/PROJECT (Minimum of 300 words)

Sankit addresses the lack of systematic production data and verifiable evidence in medicinal herb supply chains. According to the team's Market Size Research tab, Vietnam has more than 5,000 medicinal plant species, while domestic medicinal herb demand is estimated at around 80,000–100,000 tons per year, with a large share still dependent on imported materials. The same research draft also identifies more than 600 medicinal herb cooperatives nationwide. These figures suggest that the issue is not limited to individual farms; it exists across organized production groups, growing areas, and upstream supply chains. However, many domestic growing areas have not yet developed stable, standardized, and well-documented production systems.

In practice, many farms, cooperatives, and growing areas still record cultivation, harvesting, primary processing, and storage activities through paper notebooks, Excel files, photos stored separately on phones, messaging apps, and documents prepared only when an inspection or buyer request occurs. These records may support basic internal tracking, but they are often difficult to verify, difficult to connect across production stages, and difficult to reuse for buyer evaluation, GACP/Organic preparation, laboratory linkage, or supplier management. For example, a cooperative may have photos of field activities, harvest logs, drying records, and laboratory results, but these data are often not linked clearly to the correct plot, farmer, harvest batch, processing batch, sample code, and final lot. When a buyer asks for evidence about one specific batch, the cooperative may need to manually search through several disconnected sources.

This problem is more significant in medicinal herbs than in many ordinary agricultural products because medicinal herb quality is influenced by many factors before the product reaches the factory. A batch is not evaluated only by yield, appearance, or harvest date. It also depends on plant identity, seed or seedling source, growing area, soil and water conditions, microclimate, cultivation practices, pest control, harvest timing, primary processing, drying conditions, storage, and consistency across batches or seasons. WHO's guideline on Good Agricultural and Collection Practices for medicinal plants treats cultivation and collection as early stages of quality assurance for medicinal plant materials, rather than activities separated from later manufacturing (WHO GACP, 2003 – WHO: <https://www.who.int/publications/i/item/9241546271>).

In Vietnam, Circular 19/2019/TT-BYT provides the regulatory basis for Good Agricultural and Collection Practices for medicinal herbs. It creates a formal framework for cultivation, collection, processing, storage, documentation, and quality management in medicinal herb production (MOH Vietnam, 2019 – Thư Viện Pháp Luật:

<https://thuvienphapluat.vn/van-ban/The-thao-Y-te/Thong-tu-19-2019-TT-BYT-quy-din-h-ve-thuc-hanh-tot-nuoi-trong-thu-hai-duoc-lieu-422422.aspx>). In addition, Decision 1976/QĐ-TTg approved the national master plan for medicinal herb development to 2020 with orientation to 2030, showing that medicinal herb growing areas are not only a market issue but also a planned development direction in Vietnam (Prime Minister of

Vietnam, 2013 – Government Portal:

<https://vanban.chinhphu.vn/default.aspx?docid=170540&pageid=27160>).

Existing solutions such as electronic farm diaries, QR traceability systems, IoT dashboards, farm management software, and blockchain-based tools all provide partial value, but they do not fully solve the problem Sankit focuses on. An electronic diary can record activities, but it may not ensure that the required evidence is complete or connected to the right batch. A QR code can display origin information, but if the underlying field data are fragmented or self-declared, its verification value is limited. Sensors can provide environmental data, but they cannot confirm whether cultivation, harvesting, drying, storage, and approval steps were actually completed. Blockchain can protect records after they are approved, but it cannot make raw field data automatically correct. Research on digital traceability in Vietnam's agricultural value chains also shows that traceability systems depend on data quality, stakeholder participation, and practical implementation across the chain, not only on digital labels or isolated technologies (World Bank, 2023 – Digital Technology for Traceability in Vietnam's Fruit and Vegetable Value Chains:

<https://documents1.worldbank.org/curated/en/099724310182330688/pdf/IDU03bd6b7680aa0804d260a4120c0c008620286.pdf>).

Therefore, Sankit is not positioned as a standalone QR tool, farm diary, IoT dashboard, or blockchain application. It is designed as a workflow and evidence management system for medicinal herb supply chains. Its objective is to help farms, cooperatives, and enterprises organize production records into structured, reviewable, and reusable evidence.

SANKIT

Giải pháp Sản phẩm Quy trình GACP Vùng trồng Tài nguyên Về Sankit Đăng ký pilot

HỆ ĐIỀU HÀNH CHẤT LƯỢNG SỐ CHO VÙNG TRỒNG DƯỢC LIỆU

Hệ điều hành chất lượng số cho vùng trồng dược liệu

Sankit giúp chuẩn hóa quy trình, giám sát toàn diện vùng trồng, tối ưu vận hành và minh chứng sự hình thành chất lượng dược liệu bằng dữ liệu số minh bạch, đáng tin cậy.

Đăng ký pilot Xem demo

QR Plot ID Định danh lô trồng GPS + Timestamp Ghi nhận theo thời gian thực GACP Workflow Quy trình chuẩn hóa Audit-ready Report Báo cáo sẵn sàng kiểm audit

Sankit's first product package is the **GACP Readiness and Farm Trust Pack**. It includes an offline-first mobile field app, a farm or cooperative management dashboard, a workflow engine based on crop procedures and buyer requirements, a field evidence layer, batch profiles, and a report builder. Field users can scan the QR code of a plot or growing area, view assigned tasks, record completed activities, take photos or videos, capture GPS location and timestamp automatically, and synchronize data when internet connection becomes available. This design is important because many medicinal herb growing areas are located in mountainous or rural regions where connectivity and digital skills may be limited.

The management dashboard is designed for farm owners, cooperative managers, technical supervisors, and upstream enterprises. It shows growing area status, task progress, delayed activities, missing evidence, available sensor data, approval status, harvest records, processing records, and report readiness. This allows managers to detect missing information during the season rather than discovering gaps only when preparing documents for buyers, laboratories, certification partners, or inspectors.

A key design layer of Sankit is **GACP/Organic-as-Code**. Instead of treating GACP, Organic, or buyer requirements as static documents, Sankit converts them into operational workflows. The system can define what task must be done, when it should be done, who is responsible, what evidence is required, who approves the record, what alert is triggered when the task is late, and what report can be generated later. This helps transform documentation from a last-stage paperwork activity into a process embedded in day-to-day field operations.

Each medicinal herb batch can be organized into a **Quality Formation Passport**, a structured digital profile describing the batch's production history. This profile may include plant identity, plot identity, cultivation history, completed tasks, delayed tasks, missing evidence, photos or videos, sensor indicators where available, harvest information, drying or primary processing records, storage information, internal approvals, laboratory result links, and buyer-specific requirements. In this setup, QR is not the product itself; QR becomes an access point to structured batch data.

Sankit is designed around crop-specific workflows rather than one generic form for all medicinal plants. For example, Sa nhân tím has operational risks related to canopy shade, soil moisture, fruiting stage, harvest separation, and post-drying mold. Root and rhizome crops such as Ngưu tất, Tam thất nam, Đinh lăng, and Nghệ vàng require stronger control of plant identity, harvest age, washing, slicing, drying, moisture, foreign matter, and lot separation. Cà gai leo, as an above-ground medicinal herb, needs attention to cutting time, drying temperature, contamination, and moisture control. Actisô depends heavily on suitable cool-climate growing areas, seed or variety quality, and disease management. These differences mean Sankit must support crop-packs: each crop has its own task templates, required evidence, risk indicators, and report structure. This crop-pack approach is based on the team's operational research report on seven medicinal herbs.

In later stages, Sankit can develop a **Quality Trust Score** to help managers and buyers assess the completeness and consistency of production data by farm, plot, supplier, or batch. This score does not claim to measure medicinal efficacy. It focuses on operational reliability: evidence completeness, task timeliness, process adherence, deviation frequency, approval history, environmental risk, laboratory linkage, and previous lot history. For buyers, this can support supplier comparison and input risk management before final procurement decisions.

Sankit can also integrate an **AI Quality Risk Engine**. In the early stage, AI can help identify missing evidence, delayed tasks, incomplete records, abnormal sensor indicators, repeated deviations, or inconsistencies between field records and batch documents. AI does not replace agronomists, auditors, inspectors, or laboratories. It serves as a review-support and early-warning layer. When enough structured data have been collected across seasons, the system can analyze relationships between cultivation conditions, process records, and laboratory results to identify patterns that require expert review.

Another planned component is the **Verified Lab Bridge**, which links laboratory test results to the correct plot, harvest event, processing record, sample code, and batch. This is important because laboratory results lose operational value if they cannot be traced back to the exact batch and production history. Sankit may also use a **Verified Quality Ledger** to protect only reviewed and approved milestones, such as cultivation record approval, harvest confirmation, drying completion, laboratory result attachment, and batch handover. This avoids the weak claim that blockchain alone can guarantee data truth. The system treats trust as the result of workflow, evidence, review, and approval.

The expected first pilot should be narrow enough to test the system under real operating conditions. A practical pilot can focus on one medicinal herb crop or a small number of crop-packs, one organized growing area, one production season, and one buyer or upstream enterprise partner. The pilot should measure whether field users can record data with low friction; whether managers can identify missing or delayed activities; whether batch profiles improve communication with buyers; whether laboratory results can be linked correctly to production records; and whether report preparation becomes more structured. Possible success indicators include evidence completeness rate, on-time task completion rate, percentage of samples with digital chain-of-custody, buyer usage of batch profiles, and additional time required from field users.

In the long term, Sankit aims to become an operational data layer for medicinal herb supply chains. Farms and cooperatives use it to manage production workflows and records. Buyers and processors use it to monitor suppliers and reduce input risk. Laboratories and certification partners can use structured records to review documents more efficiently. Once the system is validated in medicinal herbs, the same workflow-and-evidence model can be adapted to other high-value crops that require

clear origin, process control, post-harvest documentation, and buyer-specific quality evidence.

4. BUSINESS MODEL

*Sankit is built around a **B2B/B2B2Field** model. The reason is that, in medicinal herb supply chains, the people who enter data in the field are not always the ones who receive the full economic value from that data. Farmers, field technicians, and cooperative staff may be the daily users, but the groups with clearer willingness to pay are usually cooperatives, growing-area owners, sourcing enterprises, processing factories, traditional medicine companies, functional food companies, or organizations that need to control upstream raw material quality. Therefore, Sankit does not follow a retail model targeting individual farmers. Instead, it focuses on organizations that need to manage growing areas, batch records, suppliers, and raw material risks.*

The main operational problem in the market is that medicinal herb production data are fragmented and difficult to reuse. A cooperative may have paper logs, Excel files, field photos, messages, harvest records, primary processing records, and laboratory results, but these data are often not clearly linked to the correct growing area, field worker, harvest batch, processing batch, sample code, and final lot. When working with buyers, laboratories, or when preparing GACP/Organic documentation, cooperatives often have to manually consolidate data, which increases the risk of missing evidence and makes cross-checking difficult.

For sourcing enterprises and processing factories, the problem is related to monitoring satellite growing areas and evaluating suppliers. If they rely only on paper documents, supplier declarations, or periodic inspections, it is difficult to follow the production process throughout the season and to trace issues quickly when a raw material batch has a problem. This creates the commercial basis for Sankit: the system does not only serve field data entry users, but also serves stakeholders who need production data for decision-making.

The core value of Sankit is to organize medicinal herb production data into structured records by growing area and batch. The system supports workflow tracking, field evidence, approval status, harvesting information, primary processing, storage, laboratory linkage, and reporting. As a result, data are not only stored; they can also be used for internal management, standard preparation, buyer communication, supplier monitoring, and input risk reduction.

*Sankit is not positioned as a QR tool, a farm diary app, or a standalone IoT dashboard. Sankit sells the capability to **organize, control, and reuse production data** in medicinal herb supply chains.*

Business Model Canvas

BMC Component	Sankit
Customer Segments	<i>Primary paying customers include medicinal herb cooperatives, farm owners, organized growing areas, sourcing enterprises, processing factories, traditional medicine companies, functional food companies, and organizations that need to control upstream raw material quality. Direct users include farmers, field technicians, cooperative staff, growing-area managers, and QA/QC personnel from upstream enterprises. Suitable early adopters include medicinal herb cooperatives and growing areas in Lao Cai, Ha Giang, Son La, and other regions with high-value medicinal crops, especially those that need to standardize documentation, prepare for OCOP, GACP/Organic, or supply processing enterprises.</i>
Value Propositions	<i>Sankit helps convert fragmented production data into structured records by growing area and batch. For cooperatives and growing areas, Sankit helps track field tasks, identify missing evidence, standardize documentation, and reduce the time needed to consolidate records when working with buyers or inspection parties. For sourcing enterprises and processing factories, Sankit helps monitor suppliers, review batch records, and connect data from growing areas to harvest, primary processing, and laboratory results, thereby supporting raw material risk reduction. For consulting, laboratory, or certification partners, Sankit provides more structured data, making document review more convenient than collecting information from paper logs, Excel files, scattered photos, and manual records.</i>
Channels	<i>Channel 1 – Upstream enterprises: approach sourcing enterprises, processing factories, traditional medicine companies, or functional food companies. These enterprises can require their satellite growing areas to use Sankit to synchronize production data. Channel 2 – Cooperatives and cooperative alliances: deploy through provincial cooperatives, cooperative alliances, digital training programs, or medicinal herb growing-area development programs. Channel 3 – Technical partners: cooperate with GACP/Organic consultants, laboratories, medicinal herb experts, OCOP support units, and agricultural digital transformation programs. Channel 4 – Field pilots: start with one medicinal herb crop, one growing area, one production season, and one purchasing partner to validate the value before scaling.</i>

Customer Relationships	<i>Sankit should be deployed through a guided implementation model during the first production season, rather than simply selling software and leaving customers to use it by themselves. For cooperatives and growing areas, Sankit supports process assessment, crop-pack setup, user training, data recording guidance, and issue handling during the early stage. For sourcing enterprises and processing factories, Sankit focuses on dashboard configuration, reporting, batch profile access, supplier evaluation criteria, and laboratory linkage. After the pilot, customer relationships should gradually shift toward a customer success model, including user guides, short tutorial videos, onboarding procedures, periodic support, and data-error alerts.</i>
Revenue Streams	<i>1. Initial setup fee: process assessment, growing-area configuration, account setup, crop-pack creation, QR/GPS identification, and user training. 2. Pilot implementation fee: applied to a 3–6 month pilot with one medicinal herb crop, one growing area, and one purchasing partner. 3. SaaS subscription fee: charged monthly or annually to cooperatives, farms, growing areas, or enterprises. 4. Batch/QR/batch passport fee: charged based on the number of digitized medicinal herb batches, activated QR codes, or issued batch profiles. 5. Enterprise buyer/processor package: dashboard for monitoring multiple suppliers, batch profiles, progress reports, growing-area evaluation, and laboratory linkage. 6. Advanced add-ons: Quality Trust Score, AI Quality Risk Engine, missing-record alerts, process-risk alerts, Verified Lab Bridge, and Verified Quality Ledger.</i>
Key Resources	<i>Technology: offline-first mobile application, backend, dashboard, database, QR/GPS system, photo/video storage, user permission management, data security, and workflow engine. Data: growing-area data, crop data, activity logs, photo/video evidence, harvest data, primary processing data, storage data, laboratory results, and batch-level records. Domain expertise: GACP/Organic knowledge, medicinal herb cultivation–harvesting–primary processing workflows, QA/QC, laboratory testing, and supply chain management. Market relationships: connections with cooperatives, growing areas, sourcing enterprises, processing factories, laboratories, medicinal herb experts, and consulting partners.</i>
Key Activities	<i>1. Product development: build the field app, management dashboard, workflow engine, batch profile system, reporting tools, permission system, and advanced modules. 2. Crop-pack development: convert each medicinal herb crop’s process into task templates, required evidence, risk alerts, and report formats. 3. Pilot implementation: select the medicinal herb crop, growing area, purchasing partner, configure the system, train users, monitor data throughout one production season, and measure outcomes. 4. Customer support: provide training, issue handling, data support, workflow updates, and user experience improvements for field users. 5. Partnership development: work with cooperatives, sourcing enterprises, laboratories, consulting partners, and medicinal herb growing-area development programs.</i>

Key Partners	Medicinal herb cooperatives and growing areas: pilot sites for testing workflows and collecting real operational data. Sourcing enterprises and processing factories: upstream partners that need raw material control and can encourage satellite growing areas to use the system. Laboratories: partners that support linking test results to the correct batch, growing area, sample code, and production records. Medicinal herb/agronomy experts: partners that help build crop-packs, identify crop-specific risks, and standardize required evidence. GACP/Organic/OCOP consultants: partners that support documentation requirements, report formats, and preparation workflows. Local organizations and technology partners: partners that support cooperative access, digital training, cloud infrastructure, sensors, security, and data integration.
Cost Structure	Product development costs: UX/UI design, mobile app development, backend, dashboard, database, QR/GPS system, workflow engine, reporting tools, and advanced modules. Infrastructure costs: servers, photo/video storage, security, backup, bandwidth, and system operations. Field deployment costs: growing-area assessment, travel, training, user support, process configuration, and pilot monitoring. Domain expertise costs: crop-pack development, GACP/Organic consulting, workflow standardization, report template design, and data suitability review. Customer support and market development costs: customer success, issue handling, user documentation, demos, B2B communication, and partner relationship development.

Revenue Model and Pricing Hypothesis

Sankit's revenue model is designed in multiple layers to fit different deployment stages and the paying capacity of different customer groups. In the early stage, Sankit can test three main pricing packages:

Package	Suitable Customers	Main Content	Proposed Pricing Method
Pilot Pack	One cooperative or one small growing area	Process assessment, crop-pack setup, field app, basic dashboard, training, and one-season monitoring	One-time implementation fee for 3–6 months
Cooperative Pack	Cooperatives or growing areas with multiple farmers or plots	Growing-area management, task management, evidence management, reporting, batch profiles, and QR/batch activation	Annual SaaS fee + fee per batch or activated QR code



Enterprise Buyer Pack	<i>Sourcing enterprises or processing factories</i>	<i>Multi-supplier dashboard, batch passports, reports, growing-area evaluation, and laboratory linkage</i>	<i>Enterprise subscription fee + fee based on number of suppliers or monitored batches</i>
<i>Specific pricing should not be fixed too early. The assumptions that need to be tested through pilot implementation include: whether cooperatives are willing to pay a setup fee or subscription fee; whether sourcing enterprises are willing to pay for supplier dashboards; whether field users can record data without creating too much additional workload; and whether batch profiles actually support supplier evaluation.</i> Testable Assumptions and Key Metrics <i>This model can be tested step by step instead of requiring the entire platform to be deployed from the beginning. In the first pilot, Sankit needs to test three core assumptions.</i> <i>First, field users must be able to record data with low enough friction. If the app adds too many steps, the system will be difficult to sustain even if it looks valuable on paper.</i> <i>Second, cooperative or growing-area managers must use the dashboard to identify missing records, track progress, and prepare documentation more effectively. If the dashboard does not help reduce management time or documentation errors, the value for cooperatives will be weak.</i> <i>Third, sourcing enterprises must find the batch passport and supplier dashboard useful for batch evaluation. If buyers find that batch profiles help them review raw material sources faster and in a more structured way, the B2B/B2B2Field model will have a stronger basis for scaling.</i> <i>Key metrics include: number of participating cooperatives/farms, number of active users, number of managed growing areas, number of batches with digital profiles, on-time task completion rate, evidence completeness rate, number of generated reports, number of batches linked to laboratory results, number of buyer views on batch profiles, pilot-to-paid conversion rate, average revenue per customer, and annual renewal rate.</i>			
5. MARKET FEASIBILITY The market feasibility of Sankit comes from three converging factors: the demand for medicinal herbs, Vietnam’s policy direction for medicinal herb growing regions, and the increasing need for traceability, production records, and quality evidence in agricultural and medicinal herb supply chains. Sankit does not approach the market as a general farm diary application for all farmers. It focuses on a narrower and more testable segment: medicinal herb cooperatives, organized growing areas, sourcing enterprises, processing factories, traditional medicine companies, functional food companies, and organizations that need to control upstream raw material quality.			

1. Market Context

Medicinal herbs are a relevant market for Sankit because their commercial value depends not only on production volume, but also on origin, cultivation process, harvest timing, primary processing, storage conditions, laboratory linkage, and batch-level documentation. In medicinal herb supply chains, a product may lose commercial value if the producer cannot provide clear records that connect the growing area, field activities, harvest batch, processing batch, sample code, laboratory result, and final lot.

This is consistent with the regulatory and technical direction of the sector. WHO's guideline on Good Agricultural and Collection Practices for medicinal plants treats cultivation and collection as early stages of quality assurance for medicinal plant materials, rather than activities separated from later manufacturing. Source: WHO GACP, 2003 – <https://www.who.int/publications/i/item/9241546271>.

In Vietnam, Circular 19/2019/TT-BYT provides the regulatory basis for Good Agricultural and Collection Practices for medicinal herbs. It establishes a formal framework for cultivation, collection, processing, storage, documentation, and quality management in medicinal herb production. Source: Ministry of Health of Vietnam, 2019 – <https://thuvienphapluat.vn/van-ban/The-thao-Y-te/Thong-tu-19-2019-TT-BYT-quy-dinh-ve-thuc-hanh-tot-nuoi-trong-thu-hai-duoc-lieu-422422.aspx>.

Vietnam also has a policy foundation for medicinal herb development. Decision 1976/QĐ-TTg approved the master plan for medicinal herb development to 2020 with orientation to 2030. This shows that medicinal herb growing regions are not only a market opportunity, but also part of a planned development direction. Source: Prime Minister of Vietnam, 2013 – <https://vanban.chinhphu.vn/default.aspx?docid=170540&pageid=27160>.

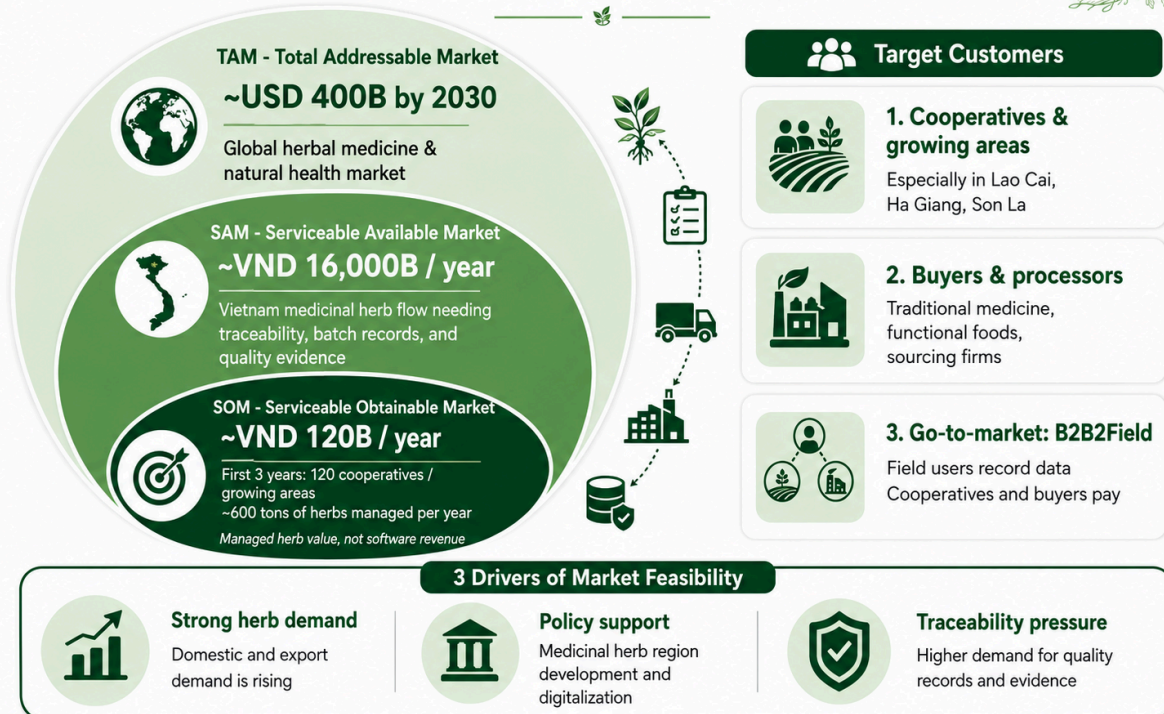
At the same time, digital traceability cannot rely only on labels or isolated technologies. A World Bank report on digital traceability in Vietnam's fruit and vegetable value chains shows that traceability depends on data quality, stakeholder participation, and implementation across the value chain, not merely on digital labels. Although the report focuses on fruit and vegetable value chains, the same logic is relevant to medicinal herb supply chains because batch records, field data, and stakeholder participation are also critical. Source: World Bank, 2023 – <https://documents1.worldbank.org/curated/en/099724310182330688/pdf/IDU03bd6b7680aa0804d260a4120c0c008620286.pdf>.

Therefore, the market opportunity for Sankit is not simply “more QR codes” or “more farm diaries.” The opportunity lies in helping medicinal herb supply chains organize production data into structured, reviewable, and reusable records that can support management, procurement, GACP/Organic preparation, laboratory linkage, and supplier evaluation.



TAM - SAM - SOM for Sankit

Market size and feasible entry scope



i SAM and SOM are sizing assumptions, not direct software revenue.

Digitizing medicinal herbs • Transparent value • Sustainable future

2. Market Sizing Logic: TAM – SAM – SOM

For preliminary market sizing, Sankit separates the value of medicinal herb flows from Sankit’s direct software revenue. This distinction is important. The value of herb batches managed through the platform is not the same as the platform’s revenue. Sankit’s revenue would come from setup fees, SaaS subscriptions, batch/QR activation, buyer dashboards, and advanced modules.

Layer	Definition in Sankit’s context	Sizing logic	Interpretation
TAM – Total Addressable Market	The broad global market related to medicinal herbs, herbal medicine, natural healthcare products, and quality/traceability infrastructure for these supply chains.	The project can use a global medicinal herb or natural healthcare market estimate as a long-term reference. If the figure of approximately 400 billion USD by 2030 is used, it should be supported by a public market report in the final submission.	TAM shows the long-term scale and global relevance of the sector. It should not be interpreted as the market Sankit can directly capture in the short term.

SAM – Serviceable Available Market	The serviceable market in Vietnam, focusing on medicinal herb production flows that require origin records, cultivation records, post-harvest documentation, laboratory linkage, and quality evidence.	As a working assumption for preliminary sizing, if Vietnam’s medicinal herb demand is estimated at 80,000–100,000 tons per year and the average herb-flow value is assumed at 200 million VND per ton , the addressable herb-flow value is approximately 16,000 billion VND per year at the lower bound of 80,000 tons.	SAM represents the value of medicinal herb batches that may need better documentation and quality evidence. It is not Sankit’s direct revenue.
SOM – Serviceable Obtainable Market	The portion Sankit can realistically approach in the first three years through pilots and early commercial deployment.	Sankit can initially target around 120 cooperatives or organized growing areas . If each unit manages around 5 tons per year , the system would manage about 600 tons per year . Using the same average value assumption of 200 million VND per ton , the herb-flow value managed through the platform would be around 120 billion VND per year .	SOM represents the value of herb batches that Sankit may digitize and manage in the first three years. Direct revenue would be a smaller software and service layer on top of this managed herb-flow value.

The purpose of this sizing is to show that Sankit operates in a supply chain where the underlying goods have meaningful value. If the platform helps reduce documentation gaps, supplier risk, and the cost of preparing quality records, customers may have a clear reason to pay even though Sankit captures only a small percentage of the total herb-flow value.

3. Target Market Segmentation

Sankit’s target market should not be defined only as “farmers.” In this sector, the more useful segmentation is based on who uses the system, who pays for it, and who benefits from the data.

Segment	Main pain point	Reason to pay or support adoption	Fit with Sankit
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Medicinal herb cooperatives and organized growing areas	Production records are often fragmented across paper logs, Excel files, photos, messages, and manual documents. It is difficult to connect records from growing area to harvest batch, processing batch, laboratory result, and final lot.	They need to standardize records, prepare documentation for buyers or standards, reduce missing evidence, and improve how they communicate production history to enterprises.	This is the most important implementation segment in the early stage because cooperatives and growing areas generate the field data.
Sourcing enterprises, processing factories, traditional medicine companies, and functional food companies	It is difficult to monitor satellite growing areas, compare suppliers, trace batch-level issues, and verify production records before procurement decisions.	They need better upstream visibility, more structured supplier data, and reduced raw material risk. They may also have stronger willingness to pay than individual farmers.	This is the strongest paying segment because these customers gain value from supplier monitoring and risk reduction.
Laboratories, consultants, certification partners, and local development programs	Records from growing areas may be incomplete, inconsistent, or difficult to review. Documentation is often prepared manually and late.	They benefit from structured data, cleaner documentation, and easier review workflows.	This segment can support adoption and credibility, but may not be the primary payer in the early stage.

The go-to-market logic should follow a **B2B2Field** model. Field workers and cooperative staff use the system every day, but cooperatives, farm owners, processors, and buyers pay because they benefit from better records, lower supplier risk, and stronger documentation readiness.

4. Geographic Focus

The initial geographic focus should be Northern Vietnam, especially provinces with medicinal herb growing potential and organized production needs such as Lao Cai, Ha Giang, Son La, and nearby mountainous or midland areas. This focus is practical for four reasons.

First, these regions have ecological conditions suitable for several high-value medicinal herbs. Many medicinal herb species require specific conditions such as altitude, cool climate, soil condition, moisture, or shade. This makes production records and crop-specific workflows more relevant.

Second, medicinal herb production in these areas often involves cooperatives, linked growing areas, or groups of farmers rather than only isolated farms. This creates a real need for a system that can manage multiple plots, users, tasks, and batches.

Third, there is a practical gap between field production and buyer requirements. Buyers and processors need more structured records, while cooperatives and growing areas often lack appropriate tools to organize those records during the season.

Fourth, the market can be tested through a narrow pilot. Sankit does not need to start nationwide. It can begin with one crop, one cooperative or organized growing area, one production season, and one buyer or processor.

5. Why the Market Needs a Specialized Solution

Medicinal herbs require a more specialized data model than many general agricultural products. A simple record of “where it was grown” and “when it was harvested” is usually not enough. Buyers may also need information about plant identity, seed or seedling source, harvest age, harvest method, drying process, moisture control, foreign matter, lot separation, storage, sample code, and laboratory result linkage.

For example, root and rhizome crops such as Nguru tất, Tam thất nam, Đinh lăng, and Nghệ vàng require stronger control over plant identity, harvest age, washing, slicing, drying, moisture, foreign matter, and lot separation. Sa nhân tím has operational risks related to canopy shade, soil moisture, fruiting stage, harvest separation, and post-drying mold. Cà gai leo, as an above-ground medicinal herb, requires attention to cutting time, drying temperature, contamination, and moisture control. Actisô depends on suitable cool-climate growing areas, seed or variety quality, and disease management.

These differences mean that Sankit should not use one generic form for all crops. The platform needs **crop-packs**, where each medicinal herb crop has its own task templates, required evidence, risk indicators, and report structure. This specialization is one reason Sankit can differentiate itself from generic farm diary or QR traceability tools.

6. Competitive and Alternative Solutions

Existing alternative	Value provided	Limitation in medicinal herb supply chains
Paper logs, Excel files, and manual documents	Easy to start, low cost, familiar to cooperatives.	Data are fragmented, difficult to verify, difficult to consolidate, and often not clearly linked across growing area, batch, sample code, and laboratory result.
Electronic farm diaries	Improve activity recording compared with paper.	Often focus on activity logs, but may not fully support field evidence, approval workflows, crop-specific requirements, batch profiles, and laboratory linkage.
QR traceability systems	Make product information easier to display to buyers or consumers.	QR is only an access point. If the underlying production data are fragmented or self-declared, verification value is limited.
IoT and sensor dashboards	Provide environmental data such as temperature, humidity, soil, or water indicators.	Sensors cannot confirm whether cultivation, harvesting, drying, storage, lot separation, and approvals were properly completed.



Blockchain or digital ledger systems	Can help protect records after they are confirmed.	Blockchain cannot make raw field data automatically correct. It still requires workflow, evidence, review, and approval before records become meaningful.
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Sankit's advantage is not any single technology. Its advantage is the combination of workflow, field evidence, batch profiles, approval status, laboratory linkage, and buyer-facing records in a system designed specifically for medicinal herb supply chains.

7. Go-to-Market Feasibility

Sankit should enter the market through a focused and testable path.

Phase 1 – Pilot with one growing area and one buyer.

The first pilot should focus on one medicinal herb crop, one cooperative or organized growing area, one production season, and one sourcing enterprise or processing partner. The objective is to test whether field users can record data with low friction, whether managers can identify missing records, whether batch profiles are useful to buyers, and whether reports help improve documentation preparation.

Phase 2 – Expand within the same linked supply chain.

If the buyer finds the data useful, Sankit can expand to other satellite growing areas or suppliers connected to the same enterprise. This reduces sales cost because the system grows through an existing buyer-supplier network.

Phase 3 – Standardize crop-packs.

After the pilot, Sankit can package crop-specific workflows into crop-packs: tasks, required evidence, risk indicators, report templates, and batch passport structure. Crop-packs make later deployments faster and more consistent.

Phase 4 – Sell enterprise buyer dashboards.

When multiple growing areas and batches are managed through the system, sourcing enterprises may need supplier dashboards, batch profiles, progress reports, laboratory linkage, and risk indicators. This is likely to become a higher-value revenue layer than selling only a field application to cooperatives.

8. Market Risks and Mitigation

Market risk	Why it matters	Mitigation
Cooperatives may be reluctant to pay for software	Many cooperatives have limited budgets and may not prioritize software spending.	Start with a focused pilot; separate setup fee, SaaS fee, and batch-based fee; sell through buyers or supported development programs where possible.

Field users may avoid data entry	If the app adds too much workload, data will become incomplete or unstable.	Use offline-first design, QR/GPS automation, short task forms, photo/video evidence, and simple onboarding.
Buyers may not pay unless they see clear value	The B2B2Field model depends on buyer or enterprise willingness to pay.	Include buyers in the pilot from the beginning; configure batch profiles and dashboards around their supplier evaluation criteria.
Input data may not be reliable enough	Poor data quality weakens reporting, Trust Score, and AI modules.	Start with simple workflows, required evidence fields, supervisor approval, and missing-record alerts before adding advanced analytics.
Existing QR or farm diary tools may be cheaper	Customers may compare Sankit with simpler tools.	Differentiate through medicinal-herb crop-packs, batch profiles, laboratory linkage, buyer dashboards, and GACP/Organic workflow design.
Field deployment costs may be high	Mountainous growing areas may require training, travel, and local support.	Start with narrow pilots; train cooperative-level champions; standardize onboarding materials and deployment process.

9. Feasibility Metrics

Sankit should measure market feasibility through adoption, operational value, buyer value, data quality, and commercial conversion.

Metric group	Specific metrics	Purpose
Adoption metrics	Number of participating cooperatives or farms; number of active users; number of managed growing areas; number of deployed crop-packs.	Measures whether the product can be introduced and used in real operations.
Operational metrics	On-time task completion rate; evidence completeness rate; number of missing records detected early; number of batches with complete profiles.	Measures whether Sankit improves production record management.
Buyer value metrics	Number of buyer views on batch profiles; number of batches reviewed through the system; buyer feedback on usefulness of batch passports.	Measures whether the system creates value for the segment with stronger willingness to pay.

Commercial metrics	Pilot-to-paid conversion rate; average revenue per cooperative or enterprise; renewal rate; deployment cost per customer.	Measures whether the model is commercially viable.
Data quality metrics	Percentage of records with GPS and timestamp; percentage of approved evidence; percentage of batches linked to laboratory results; missing-data rate by crop-pack.	Measures whether the data foundation is strong enough for Trust Score and AI modules later.

10. Conclusion on Market Feasibility

Sankit's market feasibility depends on entering the market through a focused segment rather than positioning itself as a general agricultural software product. Medicinal herb supply chains are suitable because they involve valuable raw materials, policy support for growing-region development, quality documentation requirements, and buyer-side pressure to manage suppliers and raw material risk.

In the first stage, the most feasible entry point is a medicinal herb cooperative or organized growing area in Northern Vietnam that is connected to a sourcing enterprise or processing factory. If the pilot proves that field users can record data consistently, managers can reduce documentation gaps, buyers can use batch profiles for supplier evaluation, and customers are willing to pay for setup, SaaS, or buyer dashboards, Sankit will have a practical basis for expansion.

Note: (*) Criteria are encouraged, but not mandatory.

6. INTERNAL ANALYSIS OF THE PROJECT (*)

Sankit is positioned as a workflow and production-evidence management platform for medicinal herb supply chains. The project does not mainly compete as a QR traceability tool or a basic electronic farm diary. Its focus is to organize field data, batch records, approval status, laboratory linkage, and buyer-facing documentation into a structured system.

This positioning fits the characteristics of medicinal herb production. For medicinal herbs, the value of a batch is not determined only by where it was grown or when it was harvested. It is also related to cultivation practices, collection, primary processing, storage, laboratory testing, and batch-level record traceability. Therefore, Sankit's key internal advantage is its ability to connect field-level data with the documentation, supplier-control, and raw-material evaluation needs of cooperatives, growing areas, sourcing enterprises, and processing factories.

1. SWOT Analysis

Factor	Assessment
Strengths	<i>Sankit focuses on a clear niche: medicinal herb supply chains, where origin records, production process, primary processing, testing linkage, and batch-level documentation are important. The B2B/B2B2Field model also fits the operational reality: field users enter data daily, while cooperatives, farm owners, sourcing enterprises, and processing factories receive stronger economic value. In addition, the product can be deployed in layers: starting with an offline-first app, QR/GPS identification, photo/video evidence, dashboard, and batch profiles; then expanding to Quality Trust Score, AI Quality Risk Engine, Verified Lab Bridge, and Verified Quality Ledger.</i>
Weaknesses	<i>The main weakness is field adoption risk. If the app is difficult to use, requires too many steps, or performs poorly in weak-connectivity areas, the data foundation will be weak. Medicinal herbs also cannot be managed with one generic form for all crops; each crop has different requirements for cultivation, harvesting, drying, storage, and testing. Therefore, Sankit needs crop-specific crop-packs validated by experts. Advanced modules such as AI Risk Engine or Quality Trust Score should only be deployed after sufficient structured field data have been collected.</i>
Opportunities	<i>Medicinal herb supply chains increasingly require clearer production records, traceability, and quality evidence. Sourcing enterprises and processing factories need to control raw-material regions, monitor suppliers, and reduce input risk. This creates an opportunity for buyer-led adoption, where upstream enterprises encourage their linked growing areas to use Sankit. The project can also work with laboratories, medicinal herb experts, GACP/Organic consultants, and local programs to improve credibility and reduce market-entry friction.</i>
Threats	<i>Sankit may be compared with cheaper QR tools, electronic diaries, or existing traceability platforms. If the project does not clearly explain the difference between “displaying origin information” and “managing batch-level production evidence,” it may be perceived as a more expensive alternative. Another risk is poor field data quality: records may be incomplete, late, or entered after the event. Sankit should also avoid overclaiming. The system supports the organization and review of production evidence, but it does not replace laboratory testing, certification, or professional quality assessment.</i>

2. Strategic Focus

*Based on the SWOT analysis, Sankit should focus on **production-evidence management for medicinal herb batches**, instead of positioning itself as a general traceability product.*

The first version should prioritize capabilities that can be validated directly in the field: simple mobile data entry, QR/GPS growing-area identification, photo/video evidence, task workflows, supervisor approval, management dashboard, and batch profiles.

Sankit should develop in layers. In the first stage, the project needs to prove that field data can be collected consistently and that managers or buyers can actually use the resulting records. Once the data foundation becomes complete and stable enough, advanced layers such as Quality Trust Score, AI Quality Risk Engine, Verified Lab Bridge, and Verified Quality Ledger will have a stronger basis for implementation.

The project should also begin with a narrow crop-pack. A pilot with one medicinal herb crop, one organized growing area, one production season, and one buyer will provide clearer operational evidence than a broad deployment with weak data. The first crop-pack should clearly define required tasks, required evidence, approval steps, risk indicators, and report outputs.

3. Key Risks and Control Measures

Key Risk	Impact	Control Measure
Low field adoption	<i>If users do not record data consistently, dashboards, batch profiles, and analytics will not be reliable.</i>	<i>Design an offline-first MVP with minimal steps, QR/GPS automation, photo/video evidence, and direct testing with field users.</i>
Overly generic workflow	<i>A single generic form cannot reflect the specific risks of different medicinal herbs.</i>	<i>Build crop-packs with input from medicinal herb experts, GACP/Organic consultants, and buyers.</i>
Building advanced features too early	<i>AI and Trust Score modules may consume resources before the data foundation is ready.</i>	<i>Prioritize workflow, evidence, dashboard, batch profiles, and reporting first; develop AI only after structured data are available.</i>
Unvalidated buyer value	<i>If buyers do not use batch profiles, the strongest paying segment remains unproven.</i>	<i>Include one buyer or processing factory in the pilot from the beginning and design reports around their supplier-evaluation criteria.</i>
High deployment cost	<i>If every pilot requires heavy manual support, the model will be difficult to scale.</i>	<i>Standardize onboarding, train cooperative-level champions, and reuse crop-packs for similar growing areas.</i>

4. Pilot Validation Plan

The first pilot should be designed as a controlled validation stage, involving one medicinal herb crop, one cooperative or organized growing area, one production season, and one

buyer or processing factory. The pilot will validate the core assumptions before scaling: field users can record data consistently; the system can capture GPS, timestamp, photo/video evidence, and approval status; managers can detect missing records earlier; buyers can use batch profiles for supplier evaluation; and customers have a basis for paying after the trial stage.

Validation Area	Key Metrics
Field adoption	Number of active users, task completion rate, average time per record, number of offline records successfully synchronized.
Data quality	Percentage of records with GPS/timestamp, percentage of tasks with required evidence, evidence approval rate, number of missing-record alerts.
Manager value	Dashboard usage, number of documentation gaps detected early, time required to prepare batch documentation.
Buyer value	Number of batch profiles viewed, buyer feedback, number of supplier-related questions answered using Sankit data.
Commercial feasibility	Willingness to pay, preferred pricing model, pilot-to-paid conversion rate, support cost per customer.

5. Overall Internal Assessment

Sankit has a clear internal direction because it focuses on a specific operational problem: production data in medicinal herb supply chains are often fragmented, difficult to verify, and difficult to reuse for management, procurement, and documentation. The project's strengths lie in its focused domain positioning, B2B/B2B2Field model, layered product architecture, and ability to connect field data with the needs of cooperatives, sourcing enterprises, and processing factories.

The most important condition for successful implementation is product discipline. Sankit should first prove that its core workflow works in real field conditions before expanding into AI, scoring, or ledger modules. The project's defensible value is to help organize, review, and reuse production evidence by growing area and batch. This positioning is clear enough to differentiate Sankit from basic QR tools or electronic diaries, while remaining practical enough to be validated through a pilot.

The pilot results will provide the basis for refining Sankit's workflow, standardizing crop-packs, and building a deployment roadmap for additional cooperatives, growing areas, and upstream enterprises.

7. DEVELOPMENT STRATEGY (*)

Sankit's development strategy follows a phased approach: validate the core workflow first, then standardize crop-specific deployment, and finally expand into buyer-facing and data-driven modules. This strategy is designed to avoid overbuilding advanced features before the project has reliable field data and confirmed user adoption.

Phase 1: MVP Development and Field Workflow Validation

The first development phase focuses on building a practical MVP that can operate in real field conditions. The MVP will include an offline-first mobile application, QR/GPS-based growing-area identification, task workflows, photo/video evidence, timestamped records, supervisor approval, a basic management dashboard, and batch profiles.

The objective of this phase is not to build every planned feature immediately. The priority is to test whether field users can record production data consistently, whether cooperative managers can monitor missing records and delayed tasks, and whether batch profiles can organize production evidence in a usable format.

In this phase, Sankit will focus on one medicinal herb crop, one organized growing area, one production season, and one buyer or processing partner. This controlled scope allows the team to validate the most important assumptions before scaling.

Phase 2: Crop-pack Standardization

After the first pilot, Sankit will standardize the validated workflow into a crop-pack. A crop-pack includes task templates, required evidence, approval steps, risk indicators, and report outputs for a specific medicinal herb crop.

This step is important because medicinal herbs cannot be managed with one generic form. Different crops have different requirements for cultivation, harvesting, drying, storage, and laboratory linkage. By developing crop-packs, Sankit can make future

deployments faster, more consistent, and more relevant to the actual production process of each crop.

The expected output of this phase is a reusable deployment model that can be applied to similar cooperatives or growing areas with lower setup time and lower training cost.

Phase 3: Buyer-facing Dashboard and Supply Chain Expansion

Once Sankit has stable field data and standardized crop-packs, the next step is to strengthen the buyer-facing layer. This includes supplier dashboards, batch profile access, evidence completeness indicators, laboratory linkage, and basic supplier comparison reports.

The purpose of this phase is to increase value for sourcing enterprises, processing factories, traditional medicine companies, and functional food companies. These stakeholders need better visibility into their upstream supply chains, especially when working with multiple cooperatives or growing areas. A buyer-facing dashboard helps them monitor supplier readiness, review batch records, identify missing documentation, and reduce raw material risk.

The go-to-market strategy in this phase follows a B2B2Field model. Field users continue to record production data, but cooperatives, farm owners, sourcing enterprises, and processing factories become the main paying customers because they receive direct value from better records, stronger audit readiness, and more structured supplier management.

Phase 4: Data Intelligence and Advanced Modules

Advanced modules will be developed only after Sankit has collected enough structured and reliable field data. These modules may include Quality Trust Score, AI Quality Risk Engine, Verified Lab Bridge, and Verified Quality Ledger.

The Quality Trust Score will focus on operational reliability, such as evidence completeness, task timeliness, approval history, deviation frequency, and laboratory

linkage. It will not be used to claim medicinal efficacy. The AI Quality Risk Engine will support early detection of missing records, delayed tasks, abnormal patterns, and process-risk signals. The Verified Lab Bridge will connect laboratory results to the correct growing area, harvest event, sample code, and batch. The Verified Quality Ledger will protect reviewed and approved milestones rather than raw field data.

This staged approach ensures that advanced features are built on a reliable data foundation instead of being used as unsupported claims.

Phase 5: Scaling and Partnership Development

After the workflow, crop-pack, and buyer-facing modules are validated, Sankit can expand to additional cooperatives, growing areas, and medicinal herb crops. The expansion strategy will prioritize regions with organized production, existing buyer relationships, and clear documentation needs.

Partnerships will be important for scaling. Sankit can work with medicinal herb experts, GACP/Organic consultants, laboratories, cooperatives, local development programs, and sourcing enterprises. These partners can help validate crop-specific workflows, improve credibility, support training, and reduce market-entry friction.

The long-term direction is to make Sankit an operational data layer for medicinal herb supply chains. Farms and cooperatives use it to manage field records and batch documentation. Buyers and processors use it to monitor suppliers and reduce raw material risk. Laboratories and consultants can use the structured records to support documentation review and quality-related workflows.

Development Timeline

Stage	Timeframe	Main Objectives	Key Outputs
	e		

Stage 1: MVP and pilot preparation	0–6 months	Build the offline-first app, QR/GPS identification, workflow module, evidence capture, dashboard, and batch profile structure.	MVP, pilot crop-pack draft, pilot partner selection, basic dashboard.
Stage 2: Controlled pilot	6–12 months	Test Sankit with one crop, one growing area, one production season, and one buyer or processing partner.	Field adoption data, evidence completeness data, buyer feedback, pilot report.
Stage 3: Crop-pack standardization	12–18 months	Convert validated workflows into reusable crop-packs and improve onboarding.	Standardized crop-pack, onboarding guide, deployment checklist, refined dashboard.
Stage 4: Commercial deployment	18–30 months	Expand to more cooperatives, growing areas, and buyer-linked supply chains.	Paid deployments, buyer dashboard, batch profile usage, SaaS revenue testing.
Stage 5: Advanced data modules	30–36 months	Develop Quality Trust Score, AI Quality Risk Engine, Verified Lab Bridge, and Verified Quality Ledger based on collected data.	Data-driven risk alerts, supplier analytics, lab linkage, verified milestone records.

Key Success Metrics

Sankit will measure development progress through both product and market indicators. Product indicators include active field users, task completion rate, percentage of records with GPS and timestamp, evidence completeness rate, approval rate, and number of batches with digital profiles. Market indicators include number of participating cooperatives, number of buyer or processor partners, number of batch profiles reviewed by buyers, pilot-to-paid conversion rate, and renewal potential.

These metrics allow the team to evaluate whether Sankit is only being built as a technology product or whether it is creating measurable value in real supply chain operations.

Strategic Direction

The development strategy of Sankit is intentionally narrow at the beginning. Instead of launching a broad platform for all crops and all users, Sankit will start with a controlled medicinal herb pilot, validate the workflow, standardize crop-packs, and then expand through buyer-linked supply chains. This approach reduces implementation risk, improves product relevance, and creates a stronger foundation for future AI and data-driven modules.

By following this phased strategy, Sankit can move from a field-recording MVP to a scalable quality-evidence platform for medicinal herb supply chains.

8. HUMAN RESOURCES (*)

Who is involved in the project, and what are the roles and responsibilities of each team member?

Sankit is developed by a multidisciplinary student team combining technology, data science, traditional medicine knowledge, and business development. This team structure is suitable for the project because Sankit is not only a software product; it also requires

understanding of field operations, medicinal herb production, quality documentation, and market deployment.

<i>Team Member</i>	<i>Institution / Major</i>	<i>Role in Project</i>	<i>Main Responsibilities</i>
<i>Phùng Minh Khoa</i>	<i>Ho Chi Minh City University of Technology (HCMUT) – Faculty of Mechanical Engineering</i>	<i>Team Leader / AI & IoT Lead / Product Owner</i>	<i>Leads the overall project direction, coordinates team execution, defines the product roadmap, and manages communication between technical, domain, and business workstreams. He is also responsible for the AI and IoT direction of Sankit, including data collection logic, sensor integration strategy, automation potential, and future AI-based risk detection modules.</i>
<i>Nguyễn Tấn Thành</i>	<i>Van Hien University – Computer Science Department</i>	<i>Full-stack Developer</i>	<i>Responsible for building the core software system, including the mobile/web application, backend services, database structure, user management, dashboard features, QR/GPS-based identification, and data synchronization. He ensures that Sankit can operate as a usable digital platform rather than remaining only at the</i>

				<i>concept stage.</i>
Trịnh Quốc Bảo	<i>Ho Chi Minh City University of Technology (HCMUT) – Data Science</i>	AI & Data Science Member		<i>Supports data architecture, data processing, analytics design, and AI-related modules. His responsibilities include structuring production data, designing indicators for missing records and process risks, supporting future Quality Trust Score development, and assisting in the evaluation of data quality collected from field pilots.</i>
Lê Hoàng Ngọc Xê Ri	<i>Vietnam University of Traditional Medicine – Traditional Medicine</i>	Traditional Medicine & Domain Specialist		<i>Provides domain knowledge related to medicinal herbs, traditional medicine, and quality-related requirements in herb production. As a fifth-year student specializing in traditional medicine, she helps the team validate crop-specific workflows, relevant terminology, medicinal herb handling logic, and the practical requirements behind cultivation, harvesting, primary processing, storage, and documentation. Her role is important to ensure that Sankit is not designed as a generic agriculture app, but as a system</i>

				<i>suitable for medicinal herb supply chains.</i>
Đỗ Khánh	<i>Ho Chi Minh City University of Technology (HCMUT) – Digital Business</i>	Business Development & Market Strategy Member		<i>Responsible for business model development, market feasibility analysis, customer segmentation, go-to-market strategy, pricing assumptions, and partnership planning. He helps connect the product with real market needs from cooperatives, growing areas, buyers, processors, and other stakeholders in the medicinal herb supply chain.</i>

Overall, the team is structured around five complementary capabilities. The AI and IoT direction supports automation, structured data collection, and future risk detection. The full-stack development role turns the concept into a working digital product. The data science role ensures that production records can be organized and analyzed. The traditional medicine role keeps the project aligned with the specific requirements of medicinal herb production. The digital business role connects the product to customer needs, pricing logic, and market-entry strategy.

This division of responsibilities gives Sankit a balanced execution foundation: technical development, domain validation, data strategy, and business feasibility are handled in parallel. In the first pilot, the team will work together to build the MVP, validate crop-specific workflows, test field data collection, and evaluate whether cooperatives and buyers can use Sankit's batch records and dashboard in real operations.



HIGH POTENTIAL TO CONTRIBUTE TO SOCIETY / COMMUNITY

Sankit aims to make a practical social contribution by helping small medicinal herb growing areas and cooperatives organize their production data in a clearer, more verifiable, and more usable way when working with buyers, laboratories, or quality assessment partners.

In medicinal herb supply chains, many farmers and cooperatives still rely on paper notebooks, scattered photos, messages, and Excel files to record production activities. These tools are familiar and easy to start with, but they are often not enough to prove the full production history of a medicinal herb batch. When records are missing, fragmented, or difficult to cross-check, real field work may not be properly recognized by the market.

Sankit does not claim to solve every problem in the medicinal herb sector. However, it can help address one urgent gap: turning daily production activities into structured evidence. A cultivation task, a harvest event, a drying record, a photo, a GPS location, a timestamp, and a supervisor approval can be linked to the correct growing area and batch. This gives cooperatives a clearer basis when communicating with sourcing enterprises or preparing quality-related documents.

The social impact of Sankit begins with producers. In many rural and mountainous areas, medicinal herb growers do not lack experience or effort; what they often lack is a simple tool to turn that effort into data that buyers and the market can trust. If designed for real field conditions, Sankit can support rural digital transformation in a practical way: fewer steps, offline usability, and workflows connected to daily work rather than adding another layer of paperwork.

For sourcing enterprises and processing factories, Sankit can reduce uncertainty in raw material procurement. When batch records are better organized, enterprises can identify missing information earlier, review supplier production history, and link laboratory results to the correct batch. This supports a more responsible supply chain, where procurement decisions are based not only on verbal commitments or manually prepared documents, but also on structured production evidence.

For consumers, Sankit contributes indirectly to the transparency of medicinal herb products. The project does not replace laboratory testing, certification, or professional quality assessment. Its value is to make the evidence behind those activities easier to collect, store, review, and reuse. When production data become clearer, the supply chain has a stronger foundation for building trust.

In the long term, Sankit can support community development in a practical way: helping cooperatives professionalize their records, helping small growing areas participate more confidently in quality-oriented supply chains, and helping local medicinal herb knowledge connect with data, standards, and modern markets. This is not an immediate transformation, but it is a necessary step toward a more transparent, sustainable, and fair medicinal herb sector for producers.

THE TEAM'S ASPIRATIONS AND OBJECTIVES IN SUBMITTING THE PROJECT TO THE COMPETITION

By submitting Sankit to the competition, our team hopes to validate the idea in a serious, practical, and feedback-oriented environment. Sankit comes from a specific problem in medicinal herb supply chains: production data are often fragmented, difficult to verify, and difficult to reuse for management, procurement, and quality



documentation. Therefore, our first objective is to assess whether the solution truly fits the needs of cooperatives, growing areas, and sourcing enterprises.

Through the competition, we hope to receive feedback from judges, experts, and mentors to improve the product model, business model, and pilot implementation strategy. This is an important step for Sankit to move beyond a technology concept and become a solution that can be tested in real field conditions.

After the competition, our goal is to complete the MVP, select one suitable medicinal herb crop and one growing area for pilot testing, and validate the workflow for field data recording, batch profiles, and management dashboards. In the long term, we hope Sankit can support medicinal herb cooperatives and growing areas in professionalizing production records, improving supply chain transparency, and buil